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THE EFFECTS OF 4-CHLOROPHENOXYACETIC ACID PLANT GROWTH REGULATOR ON THE FRUIT SET, YIELD AND ECONOMIC BENEFIT OF GROWING TOMATOES IN HIGH TEMPERATURES

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ABSTRACT

The effects of different concentrations of 4-chlorophenoxyacetic acid (4-CPA) plant growth regulator hormone on fruit set, yield and economic benefit of tomato (*Lycopersicon esculentum* Mill) growing in high temperatures in Botswana (Southern Africa) were examined. In a field experiment laid under complete randomized block design, tomatoes flowers were treated with four different concentrations of 00 ppm (control), 15ppm, 45ppm and 75ppm of 4-CPA growth regulator. Data collected involved number of fruit set, weight of small tomato, weight of cracked tomatoes, weight of cat face tomatoes, weight of rotten tomatoes, weight of pest damaged tomatoes and marketable tomatoes. A two way analysis of variance (ANOVA) was performed using the SPSS software ver.19 to analyze the data. The application of 4-CPA hormone indicated a positive and significant effect on the fruit set and yields of tomato. A positive relationship between the hormone concentration and the fruit set as well as total yield of tomato was also established (higher the concentration, higher the fruit set and tomato yield). The 75 ppm concentration of 4-CPA resulted not only the highest increase in fruit set but also increased the tomato yield and hence economic benefit in tomato production increased. It was concluded that use of 4-CPA hormone increased the fruit set, yield and economic benefit of summer tomato production. Suggested future research can be conducted to observe the effect of higher concentration of the 4-CPA hormone on fruit set, yield and fruit quality of tomatoes.

KEYWORDS: Plant Growth Hormone, Tomato Fruit Set, 4-Chlorophenoxyacetic Acid, Economic Benefit, Summer Tomato Production, Botswana

INTRODUCTION

Tomato (*Lycopersicon esculentum* Mill.) is an important vegetable crop and has a significant role in human nutrition in human diet Fekadu et al. (2004). Tomato is a summer crop but can be grown throughout the year. It is a major summer vegetable fruit crop in Botswana but the productivity is very poor and local demand is met by the import from South Africa. There are generally various constraints resulting in low production of vegetables including tomato in Botswana which include poor soil fertility, water scarcity, poor cultivation skills, attack of pest and disease, poor availability of inputs and harsh climate (Baliyan & Kgathi, 2009).

The harsh climate, especially extremely high temperature in summer and low in winter reduce normal vegetative and reproductive organs development for proper fruit settings and maturation in tomato. Lack of heat tolerant tomato varieties and poor fruit setting of existing varieties during the hot dry season, where the demand for tomato is very high in the local market, is one of the challenges tomato growers are facing in tomato production during summer season. Improvement in fruit setting might have important value to the farmers in increasing tomato yield. Fruit set is defined as the transition of a quiescent ovary to a rapidly growing young fruit. Tomato fruit set is very sensitive to environmental conditions, in particular, to too low or high temperatures that affect pollen development and anther dehiscence. Fruit set depends on the successful completion of pollination and fertilization (Gillaspy et al., 1993). The climate in Botswana is semi-arid and a temperature in summer exceeds $40^{o}C$. In such high temperature conditions tomato flowers often fail to set fruits which leads to low productivity of summer tomato. High temperatures reduces fruit set, fruit production and yield in tomato Peet, M.M., Willits. D. H. and Gardner, R. (1997) and Adams, S. R., Cockshull, K. E and Cave, C. R. J. (2001), in eggplant Sanwal, S.K., Baswana, K.S. and Dhingra, H.R. (1997) and also in bell pepper Ericson, A.N. and Markhart, A.H. (2001).

Plant growth regulators (PGRs) are extensively used in horticultural crops to enhance plant growth and improve yield by increasing fruit number, fruit set and size (Batlang, 2008). Plant growth regulators such as auxins and gibberellins are known to affect parthenocarpy Matlob, A.N. and Kelly, W. C. (1975), fruit setting Rappaport, L. (1957) and fruit size Osborne D.J., Went F.W. (1953); therefore synthesized auxins and gibberellins are often used for promotion of fruit set in some fruit vegetable production including tomatoes Kuo, C.G. and Tsai, C.T. (1984) and yields can increase dramatically to four times Abdulla, K.S., et al. (1978). Reports indicated increased fruit size and fruit setting in tomato due to application of plant growth regulators such as 2, 4-dichlorophenoxy acetic acid (2,4-D), 4-chlorophenoxy acetic acid (4-CPA), and β-naphthoxya cetic acid (β-NAA) (Gemici et al., 2006). Similarly, gibberellic acid (GA3) at low concentration was reported to promote fruit setting in tomato (Sasaki et al., 2005; Khan et al., 2006). Tomato plants treated with a mixture of 4-CPA and GAs (Sasaki et al., 2005) showed increased fruit set and proportion of normal fruits compared to plants of the same crop treated with 4-CPA alone. Similarly, sprays of NAA or β-NAA at the time of flowering resulted in reduced pre-harvest fruit drop and increased the number of fruits per plant (Alam and Khan, 2002).

This clearly indicates that plant growth regulators could improve tomato fruit setting and yield and might have commercial importance to the tomato growers in Botswana. However, these PGRs have not been exhaustively studied and there is lack of information regarding type and rate of application. 4-chlorophenoxyacetic acid (4-CPA) is one of the plant growth regulators which increases fruit set during summer (Sasaki et al 2005). However, No study has been conducted on the effect of 4- CPA on tomato production in Botswana.This study was therefore initiated to determine the effect of 4-Chlorophenoxy Acetic Acid (4-CPA) on the fruit set and yield of summer tomato in the climatic conditions of Botswana and also, to estimate the economic benefit of the application of the hormone.

To fulfill the objective, the null and the alternative research hypotheses were stated as under:

- $H_{01:}$ The four concentrations of hormone (treatments) treatments do not have significant effect on the total number of tomato fruit set.
- $H_{11:}$ The four concentrations of hormone (treatments) treatments have significant effect on the total number of tomato fruit set.
- $H_{02:}$ The four replications (experimental blocks) do not have significant effect on the total number of tomato fruit set.
- $H_{12:}$ The four replications (experimental blocks) have significant effect on the total number of tomato fruit set.

- $H_{03:}$ The four concentrations of hormone (treatments) treatments do not have significant effect on the total yield of tomato.
- $H_{13:}$ The four concentrations of hormone (treatments) treatments have significant effect on the total yield of tomato.
- $H_{04:}$ The four replications (experimental blocks) do not have significant effect on the total yield of tomato.
- $H_{14:}$ The four replications (experimental blocks) have significant effect on the total yield of tomato.

MATERIALS AND METHODS

The experiment was conducted at experimental field of Regional Agricultural Research Station, Maun, Botswana (Southern Africa) during the summer season. Maun was selected for this study because it experiences the extreme temperatures of -3^{0}c in winter and 44^{0}c in summer. Seeds of an indeterminate tomato variety FA 593 were planted in seed trays. The four weeks old seedlings were transplanted in the experimental plots at a spacing of 1.2 m x 40 cm. A Randomized Complete Block Design (RCBD) with four replicates was used and therefore a total of 16 plots with a size of 6 × 1.2 m each were planted under the experiment. Tomato plants were supported using strings suspended from an overhead wire and was pruned to single stem. All the recommended cultural practices were followed during the conduction of the experiment.

Four different concentrations of 00 ppm (control), 15 ppm, 45 ppm and 75 ppm (mg per litre) of 4-Chlorophenoxy Acetic Acid (plant growth hormone) were used for spray on tomato flowers. The flower clusters were sprayed only once with these concentrations of 4-CPA hormone when 3-5 flowers were opened in a cluster.

The sprayed clusters were then identified using tags of different colors so as identify the sprayed clusters and to avoid spraying again. The spray was carried out twice a week between 1400 – 1530 hours so as to realize the best results of the hormone on tomato fruit set. The number of fruit set was counted and recorded two weeks after each spray.

In order to collect data on tomato yield, the ripe tomatoes were harvested twice a week and completed in total of 12 harvesting. Harvested tomatoes were counted and graded into marketable and non marketable (very small size, cat face, cracked, and damaged by pest and disease) tomatoes. The data were analyzed using a two way analysis of variance through SPSS Software.

To compare the economic benefit of hormone use, the economics of the hormone application was estimated from the tomato production with and without hormone application. The cost of hormone use (cost of hormone and labour cost in application) was taken into consideration because the purpose was only to estimate and compare the economic benefit of hormone use and therefore total cost of tomato production was not calculated and held constant for production of tomatoes with and without the hormone application.

The cost of hormone use which included the cost of the hormone (market value of 4-CPA hormone) and cost of hormone application (labour charges for spraying hormone) therefore, the cost of hormone use was same for all the three treatments (except 00 ppm while estimating the economic benefit of hormone use in different treatments. The 00 ppm concentration of hormone (untreated) was used as a base for comparing the economic benefit of hormone use.

RESULTS AND DISCUSSIONS

The results and discussion are presented in three subheadings as follows.

Effect of Different Concentrations of 4-CPA Application on Tomato Fruit Set

A two way analysis of variance was employed to test the null hypotheses (H01 and H02) and the results are presented in Table 1. The two way analysis of variance revealed that the treatments differ significantly $\left(p-value=0.025\right)$, while there were no significant block effects $\left(p-value=0.151\right)$. In other words, the four treatments, namely, 00 PPM, 15 PPM, 45 PPM and 75 PPM contributed differently in so far as the total number of fruit set is concerned. However, the total number of fruit set did not seem to be impacted by how the experimental fields were laid out as blocks in the experimental design of the study.

Table 1: Effect of Different Concentrations of 4-CPA Application on Tomato Fruit Set

Source	SS	df	MS	F	Sig
Replications	7980.188	3	2660.063	2.253	.151
Treatments	29215.687	3	9738.562	8.248	.006
Error	10626.563	9	1180.729		
Total	**2814223.000**	**15**			

The multiple comparisons based on LSD principle revealed that the three combinations of treatments of hormone namely, (00, 75), (15, 75), (45, 75) out of the $\begin{pmatrix}4\\2\end{pmatrix}=6$ combinations were significant at 5% level. The treatment means showed that the treatment 75 PPM had the highest total number of tomato fruit set mean of 480.5, while the treatments 45, 15 and 00 turned out a mean of 422.0, 397.25 and 363.5 respectively.

The Figure 1 shows the plot of estimated marginal means of total number of fruit set over the four blocks. It was observed that the treatment 75 PPM emerged as the best treatment to be replicated in the blocks to maximize the total number of fruit set, while the treatment 00 PPM produced the lowest total number of tomato fruit set among the four treatments of 4-CPA hormone. As expected the treatment 00 PPM produced the lowest fruit set because it was the CONTROL where no hormone was applied. It simply reflected that application of hormone not only improved the fruit set but also fruit set increased as the concentration of hormone increased and therefore, a positive relationship between the concentration of hormone and the number of fruit set was also established.

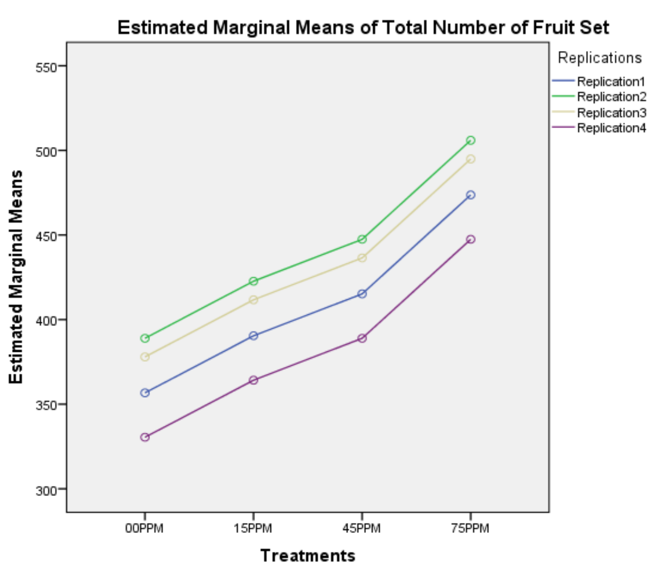

Figure 1: Estimated Marginal Means of Total Number of Tomato Fruit Set

The number of fruit set was recorded just before the commencement of harvest and was done for each treatment and spraying dates separately. Application of 75 ppm concentration of 4-CPA hormone produced the highest number of tomato fruit set (480.5) followed by 422 at 45 ppm concentration. The lowest number of tomato fruit set (363.5) was observed in control where no hormone was applied. The tomato fruit set was increased by 9.28%, 16.09% and 32.19% at 15, 45 and 75 ppm concentrations of hormone respectively, as compared to the fruit set by the treatment plot where no hormone was applied. It also indicated that the number of fruit set has increased with the increase in the concentration of hormone applied. Ozguven, A.I. and others 1997 found similar results where a positive relationship between the fruit set and the concentration of hormone was observed. Similar results were also observed by Sasaki, H and others (2005) in their field experiment where tomatoes treated with 4-CPA showed increase in fruit set. It is therefore clear that the application of 4-CPA hormone has significant effect on the tomato fruit set.

Effect of Different Concentrations of 4-CPA Application on Tomato Yield

A two way analysis of variance was employed to test the null hypotheses (H_{03} and H_{04}) and the results were presented in Table 2. Table 2 reflects that the treatments differ significantly $\left(p-value=0.009\right)$, while there are no significant block effects $\left(p-value=0.651\right)$. In other words, the four treatment combinations, namely, 00 PPM, 15 PPM, 45 PPM and 75 PPM contribute differently in so far as the total yield of tomato is concerned. However, the total yield does not seem to be impacted by how the experimental fields are laid out as blocks in the study design.

Table 2: Effect of Different Concentrations of 4-CPA Application on Tomato Yield

Source	SS	df	MS	F	Sig
Replications	132.086	3	44.029	0.567	0.651
Treatments	1681.498	3	560.499	7.214	0.009
Error	699.305	9	77.701		
Corrected Total	**2512.889**	**15**			

The multiple comparisons based on LSD principle reveal that the following three combinations of treatments, namely, (00, 75), (15, 75), (45, 75) out of the $\binom{4}{2}=6$ combinations are significant at 5% level. Going by the treatment means it is clear that the treatment 75 PPM has the highest tomato yield of 72.195 kgs, while the treatments 15, 45 and 00 turned out a total yield of 53.410, 58.215 and 43.765 kgs respectively. The Figure 2 shows the plot of estimated marginal means of total tomato yield over the four blocks of experiment. It can be seen that the treatment 75 ppm hormone concentration emerges as the best treatment to be replicated in the blocks to maximize the total tomato yield output, while the treatment 00ppm concentration corresponds to the lowest total yield among the four treatments.

The total yield of tomato was the highest of 72.20 kg and 58.21 kg at 75 ppm and 45 concentration of hormone application, respectively followed by 15 ppm (53.41 kg). The lowest number of tomato fruit set (363.5) was observed in control where no hormone was applied. The tomato yield was increased by 22.05%, 33.02% and 64.99% at 15, 45 and 75 ppm concentrations of hormone, respectively as compared to the yield produced by the treatment plots where no hormone was applied (00ppm). It also indicated that the tomato yield has increased with the increase in the concentration of hormone applied. It therefore clearly indicates that the application of 4-CPA hormone has positive effect on yield of tomato growing in higher temperatures. Results also revealed that tomato yield has increased with the increase in the concentration of 4-CPA hormone.

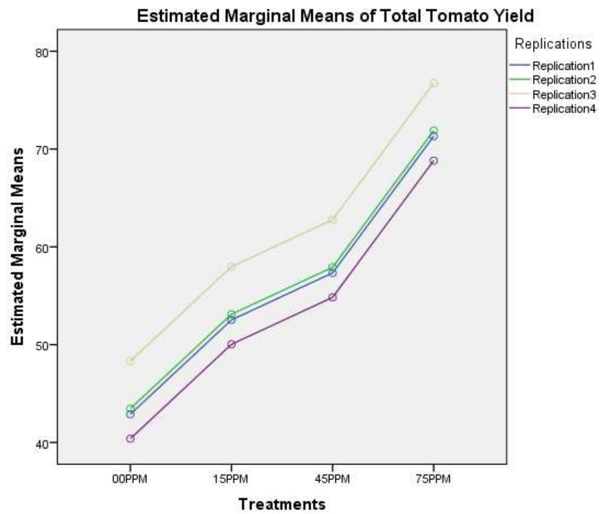

Figure 2: Estimated Marginal Means of Tomato Yield

A few secondary hypotheses of research interest in this study were to find out whether four treatments of hormone concentrations have any adverse effect on pertinent characteristics related to tomato yield. These characteristics were (i) weight of small tomatoes, (ii) weight of cracked tomatoes, (iii) weight of cat face tomatoes (vi) weight of rotten tomatoes and (v) weight of pest damaged tomatoes. In order to examine these, one way ANOVA was performed by considering the four treatments as factors and the weight of tomatoes under each of the five test characteristics as the dependant variable. Interestingly, none of the four treatments of hormone concentration seemed to have any adverse impact on the aforementioned test characteristics of tomato yield. The corresponding $p-values$ for weight of small tomato, weight of cracked tomatoes, weight of cat face tomatoes, weight of rotten tomatoes and weight of pest damaged tomatoes were 0.441, 0.352, 0.683, 0.456, 0.819, respectively.

Economic Benefit of Hormone Application in Tomato Production

The estimated economic benefit of different concentrations of hormone application is presented in Table 3.

Table 3: Estimated Economic Benefit of Hormone Application in Tomato Production

Treatment	Returns (BWP/ha)	Cost* (BWP/ha)**	Benefit (BWP/ha)	Benefit Increase (%)
00 ppm	154983	0	154983	0
15 ppm	187707	5245	182462	17.73
45 ppm	206160	5245	200915	29.64
75 ppm	255705	5245	250460	61.6

* Cost refers to cost of hormone and its application cost only
** BWP refers to the currency of Republic of Botswana called "Botswana Pula"

The application of 75 ppm concentration of hormone application increased the highest returns by 61.60% followed by 45 ppm treatment 29.64% and 17.73, respectively. The results indicated that there is a positive relationship between the increase in benefit of hormone use and the hormone concentration. Therefore, the results have established that use of hormone on summer tomato production has economic benefit in tomato production.

CONCLUSIONS AND RECOMMENDATIONS

The application of 4-CPA hormone has positive and significant effect on the fruit set and total yield of summer tomato. A positive relationship between the concentration of hormone and the number of fruit set as well as tomato yield (higher the concentration higher the fruit set and yield was observed. It was found that 75 ppm concentration of 4-CPA resulted not only the highest increase in fruit set (32.19%) but also increased the tomato yield by 64.99%. Thus, hormone application have resulted in significantly increase in number of tomato fruit set which have obviously resulted in significant increase in tomato yield. Observing the economic benefit of hormone use on summer tomato production, it was concluded that all the concentrations of hormone yielded into reasonable benefit increase by 61.60, 29.64 and 17.73%, respectively as compared to the yield produced by the tomato where no hormone was used (00 ppm). A positive relationship between the hormone concentration and the economic benefit was also established. Considering the positive effect of 4- CPA hormone on fruit set, tomato yield and the economic benefit, it can be concluded that use of 4-CPA hormone has a great potential to increase the summer tomato yield which can increase the farm income. This study provides a strong possibility that higher concentration of 4-CPA (>75ppm) may have more positive effects on the yield of summer tomato and therefore it was suggested that similar study can be conducted to determine the effects of higher concentration of 4-CPA hormone on fruit set and yield of tomato.

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